



OTTO VON GUERICKE
UNIVERSITÄT
MAGDEBURG

MAGDEBURG RESEARCH AND COMPETENCE CLUSTER VLBA

ARBEITSGRUPPE WIRTSCHAFTSINFORMATIK



Anomaly Detection in Manufacturing

VLBAII – System Architectures

Magdeburg Research and Competence Cluster
Workgroup Business Informatics I

Faculty of Computer Science
Department of Technical and Business Information Systems

Otto-von-Guericke-University Magdeburg

Anjali Kalwar

Rohith Nukala

Sourya Shome

Waseem Ashraf Khan

Business Problem & Objectives

Business Problem:

- ❖ Rise in machine breakdowns of a PCB manufacturing company which causes downtime and leads to high added costs.

Business Objectives

- ❖ Improve production efficiency
- ❖ Reduce unplanned downtime
- ❖ Detect breakdowns early
- ❖ Reduce maintenance and repair cost
- ❖ Provide clear insights into key factors causing machine wear.

Project Team Composition

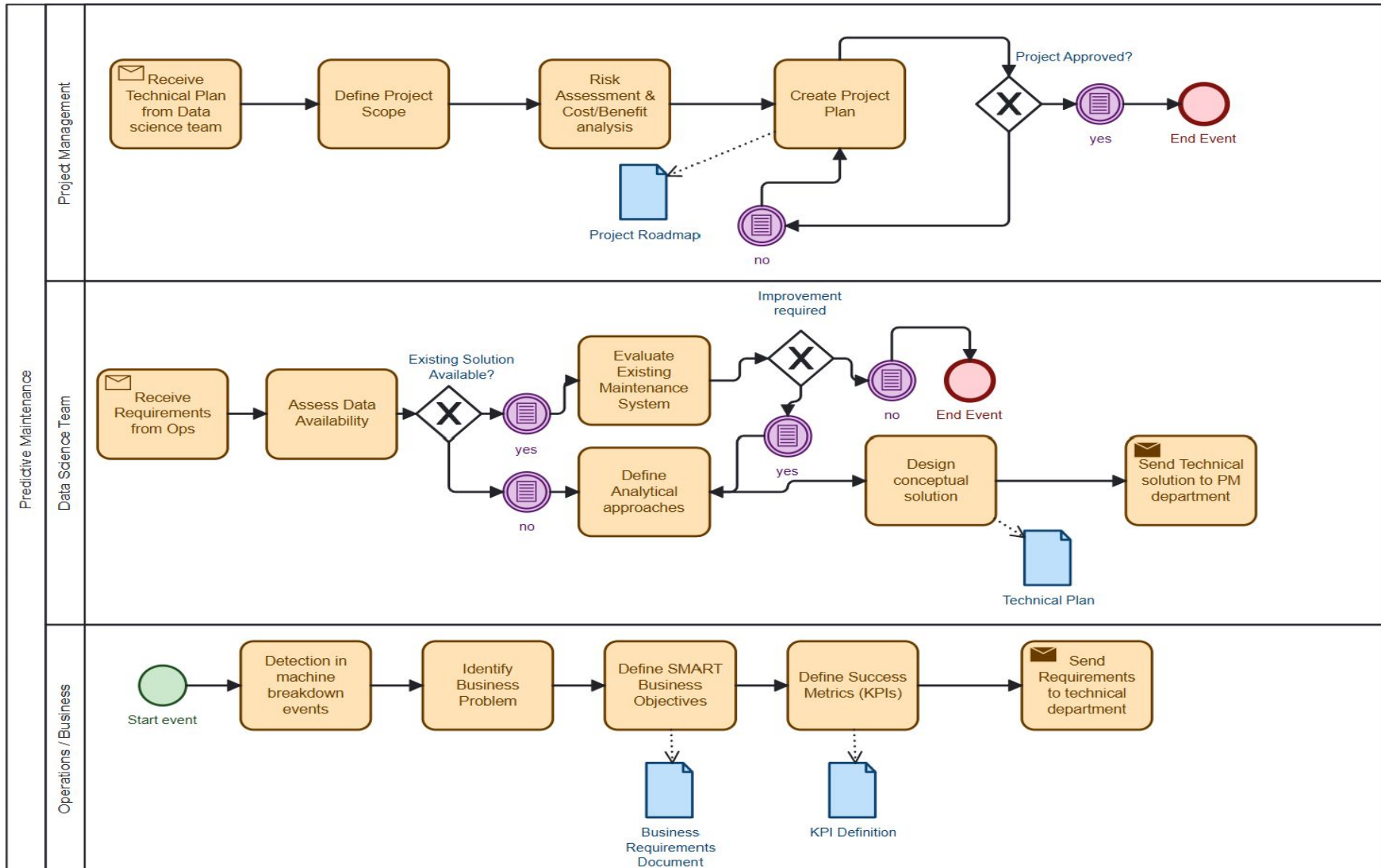
Strategy & Project Management: Project management aspects	Sourya Shome Waseem Ahsraf Khan Rohith Nukala
Data Management & Analysis: data collection, exploration, and preparation	Sourya Shome Waseem Ashraf Khan Anjali Kalwar
Analytical Modeling: building and evaluating (ML) models	Sourya Shome Waseem Ashraf Khan Anjali Kalwar
Data Infrastructure & Operations: responsible for technological infrastructure for model deployment, maintenance, and scaling	Sourya Shome Anjali Kalwar Rohith Nukula



Project
coordination: **Notion**

Documentation space:
OVGU cloud

Conceptual Diagram



Conceptual diagram(BPMN)

Project Goal

Project goal:

- ❖ Implement an ML-based anomaly detection system to predict machine failures leveraging MLOPS and support predictive maintenance.



Requirements

Functional Requirements:

- ❖ Production data acquisition.
- ❖ Detection of potential machine breakdowns.
- ❖ Provision of actionable maintenance alerts.

Non Functional Requirements:

- ❖ Explainability of prediction results.
- ❖ Transparency and traceability of MLOps lifecycle.
- ❖ Reproducibility and Continuous monitoring of data drift.

Business Architecture for Predictive Maintenance

Proposed solution:

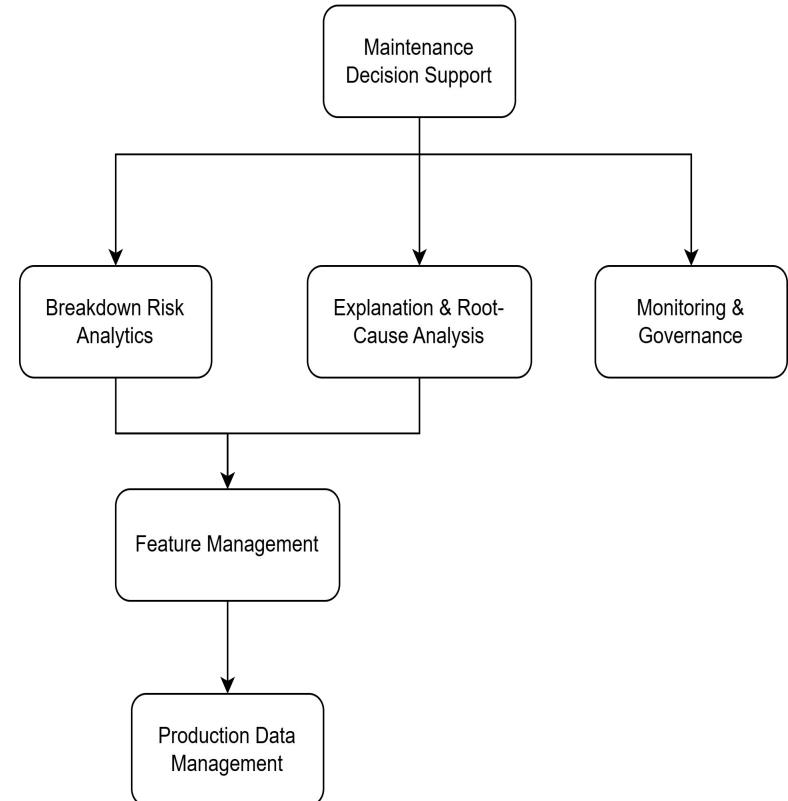
1. End-to-end MLOps-based predictive maintenance pipeline from raw production data to maintenance alerts.
2. Using a layered architecture with data ingestion, feature engineering, model training, evaluation, inference, monitoring and dashboarding.

System Input:

- ❖ Job-level schedule data including priority, family type, stage machine IDs, processing times, waiting time, tardiness, and BREAKS

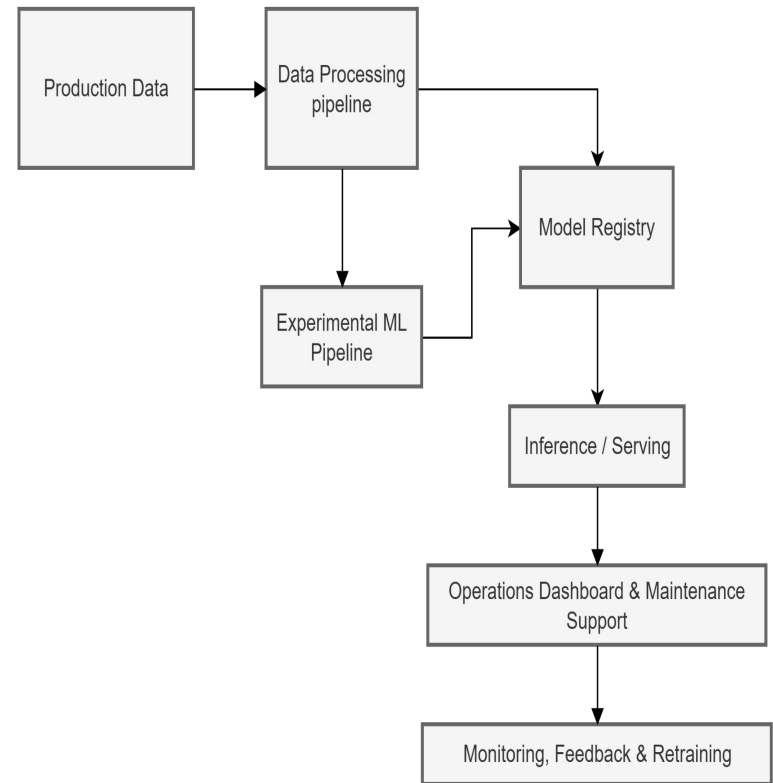
System Output:

- ❖ Breakdown risk score for jobs or machine-stage combinations.
- ❖ Explainable alert with likely wear-driving factors.
- ❖ Decision support for maintenance planning



Technical Plan

- ❖ Data integration layer – connect relevant production and machine data sources to the analytics workflow.
- ❖ Data preparation pipeline – establish a reusable pipeline for validation, transformation, and provision of analysis-ready data.
- ❖ Model development environment – develop and evaluate the anomaly detection model in an experimental setup.
- ❖ Deployment architecture – prepare a serving component that delivers predictions to business users.
- ❖ Monitoring and maintenance – monitor model and system performance and enable continuous improvement over time.

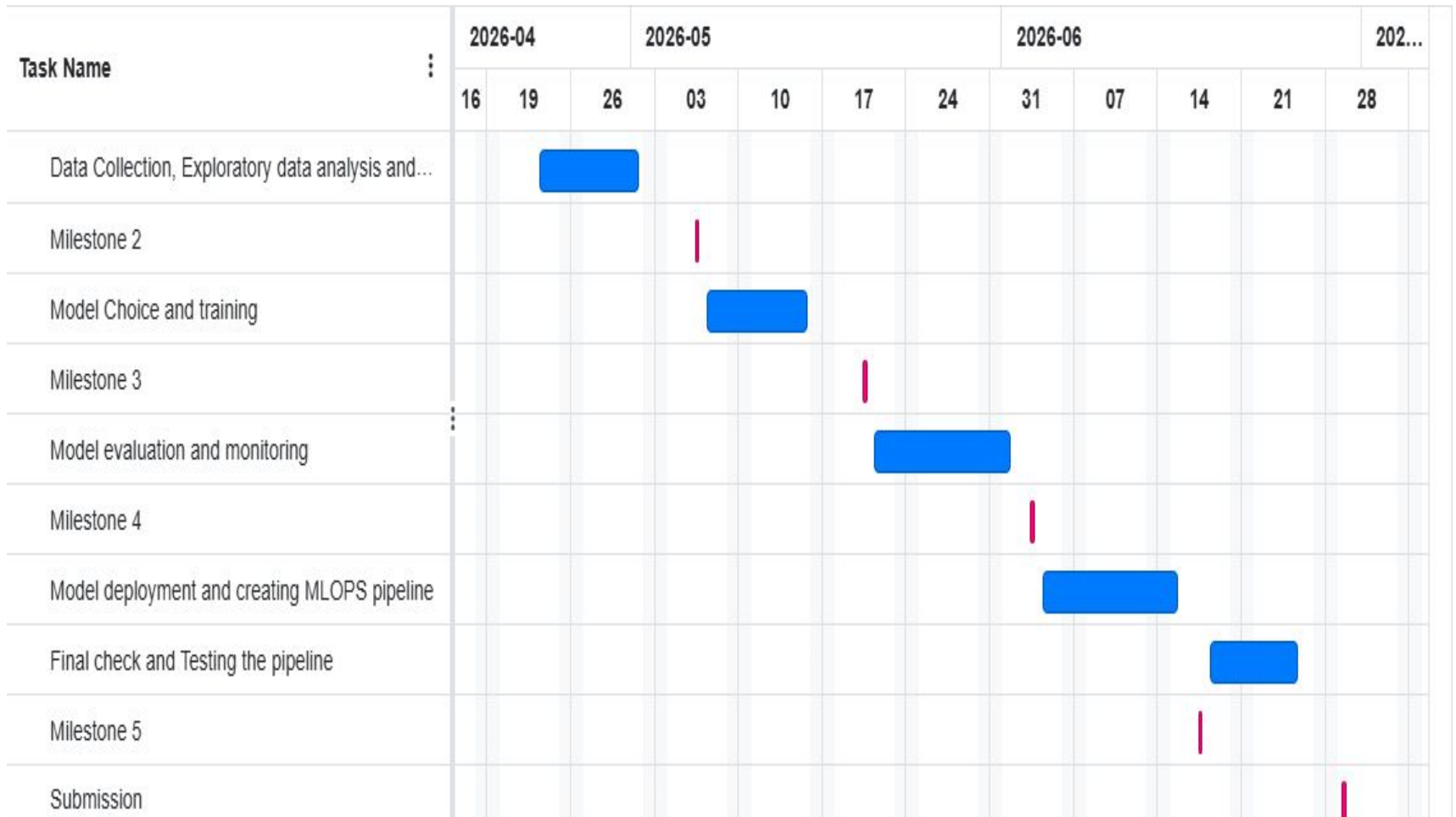


Workflow Diagram

Success Metrics

- ❖ Provide early warnings and maintain consistent performance for effective maintenance decisions.
- ❖ **High recall** to correctly detect breakdowns.
- ❖ Ensure **high precision** to reduce false alarms and avoid unnecessary maintenance actions.
- ❖ F1-score to balance false alarms, supporting overall reliable and cost-effective maintenance planning

Gantt Chart



Thank you